Manifold Learning Based on Locally Linear Embedding for Symmetric Positive Definite Matrix☆

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Abstract

Symmetric positive definite (SPD) matrices have been widely utilized as feature descriptors in machine learning. To alleviate the computational burden associated with the high dimensionality of the original SPD matrices, dimensionality reduction (DR) techniques have been employed for SPD matrices extensively. However, classic manifold learning algorithms for DR are primarily designed for vector-based data. To address this issue, this paper proposes a novel DR algorithm for SPD matrices named Lie-LLE, utilizing the basic idea of the popular manifold learning algorithm locally linear embedding (LLE) and leveraging the favorable properties of the Lie group structure inherent in the SPD matrix manifold. Specifically, we employ LLE for DR in the tangent space without vectorizing matrices, and obtain low-dimensional SPD matrices via the exponential map. Experiments conducted on several

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visualization and image set classification tasks demonstrate that Lie-LLE exhibits superior performance compared to other advanced DR algorithms for SPD matrices.
Keywords: Manifold learning, Symmetric positive definite, Locally linear embedding, Log-Euclidean metric, Image set classification

1. Introduction

The manifold of symmetric positive definite (SPD) matrices is a significant research subject at the intersection of differential geometry and data science. A set of $n \times n$ SPD matrices possesses a smooth manifold structure[1], on which various Riemannian metrics can be defined (e.g., Affine Invariant Riemannian metric (AIRM)[2], Log-Euclidean metric (LEM)[3]). Based on its unique geometric properties, the SPD matrix manifold demonstrates significant application value across multiple fields. Specifically in computer vision, covariance descriptors can naturally integrate the statistical information of regional features (e.g., color, gradient), which are widely employed in pedestrian detection[4, 5], image classification[6, 7] and action recognition tasks[8, 9]. In the field of signal processing, covariance matrices characterize radar signal information for target detection[10, 11]. Within biomedical research, SPD-based representations of medical signals and images have demonstrated efficacy in disease diagnosis and pathological pattern analysis[12, 13, 14]. Overall, these applications highlight that SPD matrices, typified by covariance matrices, provide a natural framework for the geometric representation of high-dimensional statistical data.

However, with the increasing dimensionality of data, the high dimension-

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ality of SPD matrices poses substantial challenges to computational efficiency and accuracy in machine learning. The DR problem on the SPD matrix manifold has attracted significant interest.

The DR of SPD matrices typically yields results in three forms. The first form converts SPD matrices into low-dimensional vectors. For example, Wang et al.[15] utilized LEM to construct a kernel function that maps SPD matrices to a high-dimensional Euclidean space, enabling the application of Euclidean-based algorithms through the kernel trick. Similarly, Xu[16] employed polynomial kernel mappings to project SPD matrices into a higher-dimensional space and derived low-dimensional vector representations using the polynomial mapping algorithm.

The second form, based on metric learning, focuses on preserving pairwise distances rather than generating explicit low-dimensional embeddings. For instance, Huang et al.[17] used LEM to learn a more discriminative tangent space for SPD matrix manifold and obtained low-dimensional distances by optimizing a Mahalanobis-like matrix.

The third form, generates low-dimensional SPD matrices, typically derived through a bilinear mapping. Most methods in this category leverage local affinity. These approaches incorporate class label information by defining intra-class attraction and inter-class repulsion coefficients, constructing an objective function optimized to enhance intra-class compactness and inter-class separation after DR process. The Riemannian conjugate gradient method is commonly employed to solve for local optima in these formulations. Specifically, Harandi et al.[18] measured SPD matrix similarity using AIRM and other divergences, while Gao et al. [19] and Xu et al.[20] utilized

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LEM. Ren et al.[21] constructed the affinity coefficients based on Gaussian distribution distances. However, these supervised DR methods prioritize class discrimination over preserving the intrinsic geometric structure of data. Overemphasis on class boundaries may inadvertently distort the underlying nonlinear manifold structure.

To overcome this limitation, certain DR methods rooted in manifold learning algorithms learns a bilinear mapping that can preserve the geometric features of data. For instance, some scholars use the two-directional two-dimensional PCA ($(2D)^2$PCA) proposed by Zhang et al.[22] to generate low-dimensional SPD matrices. Nevertheless, this global optimization approach may overlook local geometric nuances. Li et al.[23] addressed this limitation by introducing Lie-LPP, which adapts the locality preserving projection (LPP) algorithm[24] to construct a Laplacian matrix within the tangent space of SPD matrices. This innovation provides a novel framework for DR on SPD manifold, enabling the direct application of classical manifold learning techniques in tangent space to derive low-dimensional matrices that retain intrinsic manifold structures.

Inspired by the fact that the SPD matrix manifold is locally isometric to its tangent space under the LEM, we propose a DR algorithm based on the popular manifold learning algorithm locally linear embedding (LLE)[25] for SPD matrix, called Lie-LLE. Similar to Lie-LPP, our approach conducts DR in the tangent space of SPD matrix manifold, capitalizing on the computational efficiency inherent in the tangent space framework. However, unlike Lie-LPP, which indirectly preserves local geometric relationships through graph embedding, Lie-LLE directly captures the nonlinear manifold struc-

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ture of data via local linear reconstruction, thus exhibiting superior expressive power for complex nonlinear manifolds.

In summary, the main contributions of this paper are given as follows:

- To overcome the limitation of LLE that can only handle vector-form data, we apply LLE to the SPD matrix manifold and proposed a novel DR algorithm for SPD matrices.
- To preserve the nonlinear manifold structure of the SPD matrices, we utilize the idea of LLE to construct a local linear reconstruction model in the tangent space, achieving geometric consistency in low-dimensional space by minimizing the reconstruction errors.
- To enable the model trained by LLE to be applicable to new samples, we learn a mapping function that can directly map subsequent high-dimensional matrices to low-dimensional ones, extending the theory and broadening the application of LLE.

The remainder of this paper is organized as follows. Section2 gives some notations that are often used in this paper. Section3 systematically reviews the SPD matrix manifold and the manifold learning algorithm LLE as the theoretical foundation for the proposed approach. Section4 elaborates the LLE framework formulated on SPD matrices through rigorous mathematical derivation. Section5 systematically presents and analyzes experimental results to validate the efficacy of the proposed method. Finally, Section6 summarizes concluding remarks and future research directions.

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2. Notations

In this paper, bold capital letters denote matrices (e.g., $\boldsymbol{X}_1$) and bold lowercase letters denote column vectors (e.g., $\boldsymbol{x}_1$). $\boldsymbol{I}_D$ is the $D \times D$ identity matrix. S_+^D denotes the SPD matrix manifold spanned by $D \times D$ real SPD matrices with dimensionality of $D(D+1)/2$. S^D denotes the space spanned by $D \times D$ symmetric matrices, which is the tangent space of S_+^D. $\mathbf{0}_{D\times D}$ and $\mathbf{0}_D$ respectively represent a $D \times D$ matrix or a $D \times 1$ vector where all elements are zeros. $\text{Diag}(\lambda_1, \cdots, \lambda_N)$ is a diagonal matrix with the diagonal elements $\lambda_1, \cdots, \lambda_N$. The principal logarithm $\log(\cdot) : S_+^D \to S^D$ is defined as

$$\begin{aligned} \log(\boldsymbol{X}_1) &= \sum_{r=1}^{\infty} \frac{(-1)^{r-1}}{r} (\boldsymbol{X}_1 - \boldsymbol{I}_D)^r \\ &= \boldsymbol{U} \,\text{Diag}(\log(\lambda_1), \cdots, \log(\lambda_N)) \boldsymbol{U}^T, \end{aligned} \tag{1}$$

where $\boldsymbol{X}_1 \in S_+^D$ and $\boldsymbol{X}_1 = \boldsymbol{U} \,\text{Diag}(\lambda_1, \cdots, \lambda_N)\boldsymbol{U}^T$ is the eigenvalue decomposition of $\boldsymbol{X}_1$. Similarly, the matrix exponential $\exp(\cdot) : S^D \to S_+^D$ is defined as

$$\begin{aligned} \exp(\boldsymbol{V}_1) &= \sum_{r=0}^{\infty} \frac{1}{r!} (\boldsymbol{V}_1)^r \\ &= \boldsymbol{U} \,\text{Diag}(\exp(\lambda_1), \cdots, \exp(\lambda_N)) \boldsymbol{U}^T, \end{aligned} \tag{2}$$

where $\boldsymbol{V}_1 \in S^D$ and $\boldsymbol{V}_1 = \boldsymbol{U} \,\text{Diag}(\lambda_1, \cdots, \lambda_N)\boldsymbol{U}^T$.

3. Background

In this section, the mathematical foundations of SPD matrix manifold are systematically introduced, including its definition and geometric properties. Subsequently, the key steps of the classic manifold learning algorithm LLE is reviewed as the theoretical foundation for the proposed method.

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3.1. The manifold of SPD matrix

A real SPD matrix $\boldsymbol{X}_1$ with size $D \times D$ satisfies $\boldsymbol{v}^T \boldsymbol{X}_1 \boldsymbol{v} > 0$ for all non-zero $\boldsymbol{v} \in R^D$. The set of $D \times D$ SPD matrices does not form a linear space under the matrix addition and multiplication. Consequently, the Euclidean metric (i.e., Frobenius norm) is inadequate for capturing the intrinsic geometry of the SPD matrix manifold, which may lead to the swelling effect on tensors[26] (i.e., the determinant of the Euclidean mean of tensors can be strictly larger than the determinants of the original tensors). In fact, when S_+^D is equipped with an appropriate Riemannian metric, it forms a specific Riemannian manifold, known as the SPD matrix manifold. Over the years, many metrics have been proposed for the SPD matrix manifold. For example, Pennec et al.[2] introduced the AIRM, which accurately represents the underlying structure of the SPD matrix manifold and eliminates the swelling effect. However, its advantages come with high computational complexity. Later, Arsigny et al.[3] proposed the LEM, which is more concise and easier to compute than AIRM while conserving the same excellent theoretical properties. Moreover, several other representative measures, such as Bregman divergence[27], Stein divergence[28], and Jeffrey KL divergence[18], can also be used to characterize the geometric disparities.

Within this study, the LEM metric, which is rooted in the Lie group structure of the SPD matrix manifold, is employed to quantify the geometric distance between arbitrary SPD matrices. To understand the origin of LEM, the foundational Lie group properties of SPD matrices are briefly reviewed as follows.

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Let $\boldsymbol{X}_1, \boldsymbol{X}_2 \in S_+^D$, their logarithmic product $\boldsymbol{X}_1 \odot \boldsymbol{X}_2$ is defined by

$$\boldsymbol{X}_1 \odot \boldsymbol{X}_2 := \exp(\log(\boldsymbol{X}_1) + \log(\boldsymbol{X}_2)). \tag{3}$$

With the logarithmic product, the SPD matrix manifold S_+^D is a Lie group[3]. Moreover, the geodesic distance between two SPD matrices $\boldsymbol{X}_1, \boldsymbol{X}_2 \in S_+^D$ under the LEM is given as[3]

$$\begin{aligned} d(\boldsymbol{X}_1, \boldsymbol{X}_2) =& \| \log_{\boldsymbol{X}_1}(\boldsymbol{X}_2) \|_{\boldsymbol{X}_1} \\ =& \| \log(\boldsymbol{X}_2) - \log(\boldsymbol{X}_1) \|_{\boldsymbol{I}_D}. \end{aligned} \tag{4}$$

It further establishes that the SPD matrix manifold is locally isometric to its tangent space under the LEM, a property which is the fundamental premise of our approach.

3.2. Locally linear embedding (LLE)

The manifold learning algorithm LLE is a classical nonlinear DR technique for vector-form data points. It assumes that the data is sampled from an underlying manifold which is locally flat (i.e., the data points on the manifold can be represented locally and linearly by their nearest neighbors). LLE generates the low-dimensional representation that maintains the linear relationships of the high-dimensional data, recovering the global nonlinear structure from the local linear fits. This clear mathematical framework has significantly advanced manifold learning research, inspiring derivative methods such as Hessian locally linear embedding (HLLE)[29] and local tangent space alignment (LTSA)[30]. Given N input data points $\{\boldsymbol{x}_1, \cdots, \boldsymbol{x}_N\}$ with $\boldsymbol{x}_i \in R^D$, the main steps of LLE are given as follows:

- Adjacency graph construction

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For each data point $\boldsymbol{x}_i$, its k-nearest neighbors $\boldsymbol{x}_j (j \in \mathcal{O}(i))$ are identified using the k-nearest neighbor algorithm, where $\mathcal{O}(i)$ denotes the nearest neighbor indices set of the ith point.

- Local linear reconstruction

The weight $\boldsymbol{W}_{ij} (j \in \mathcal{O}(i))$ that reconstructs $\boldsymbol{x}_i$ is computed by minimizing the cost function

$$f(\boldsymbol{W}) = \sum_{i=1}^{N} \left\| \boldsymbol{x}_i - \sum_{j \in \mathcal{O}(i)} \boldsymbol{W}_{ij} \boldsymbol{x}_j \right\|^2, \tag{5}$$

$$\text{s. t.} \sum_{j \in \mathcal{O}(i)} \boldsymbol{W}_{ij} = 1. \tag{6}$$

- Low-dimensional embedding

The low-dimensional embedding vector $\boldsymbol{y}_i$ is reconstructed by $\boldsymbol{W}_{ij}$. This is achieved by minimizing the cost function:

$$\psi(\widetilde{\boldsymbol{Y}}) = \sum_{i=1}^{N} \left\| \boldsymbol{y}_i - \sum_{j \in \mathcal{O}(i)} \boldsymbol{W}_{ij} \boldsymbol{y}_j \right\|^2, \tag{7}$$

where $\widetilde{\boldsymbol{Y}} = [\boldsymbol{y}_1, \cdots, \boldsymbol{y}_N]$. In order to obtain the standardized low-dimensional representations, $\widetilde{\boldsymbol{Y}}$ should satisfy

$$\sum_{i=1}^{N} \boldsymbol{y}_i = \boldsymbol{0}_d, \frac{1}{N} \sum_{i=1}^{N} \boldsymbol{y}_i \boldsymbol{y}_i^T = \boldsymbol{I}_d. \tag{8}$$

The Lagrange multiplier method can be used to solve this constrained optimization problem. After simple algebraic operations, it can be reduced to a eigenfunction problem as

$$\boldsymbol{M} \widetilde{\boldsymbol{Y}}^T = \lambda \widetilde{\boldsymbol{Y}}^T, \tag{9}$$

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where $\boldsymbol{M} = (\boldsymbol{I} - \boldsymbol{W})(\boldsymbol{I} - \boldsymbol{W})^T$, and the weight matrix $\boldsymbol{W}_{k\times N}$ is extended to $\boldsymbol{W}_{N\times N}$ filling excess elements with zeros, i.e., $(\boldsymbol{W}_{N\times N})_{ij} = (\boldsymbol{W}_{k\times N})_{ij}$ if $j \in \mathcal{O}(i)$, otherwise $(\boldsymbol{W}_{N\times N})_{ij} = 0$. After obtaining the smallest $d+1$ eigenvalues $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_{d+1}$ and the corresponding eigenvectors $\boldsymbol{a}_1, \boldsymbol{a}_2, \cdots, \boldsymbol{a}_{d+1}$, the low-dimensional representation $\widetilde{\boldsymbol{Y}}$ is usually determined by $[\boldsymbol{a}_2, \cdots, \boldsymbol{a}_{d+1}]^T$, which discards the bottom eigenvector. This is because $\boldsymbol{M}$ is a positive semi-definite sparse matrix whose smallest eigenvalue always close to 0 with a unit eigenvector, which does not capture the intrinsic geometry of manifolds.

4. Proposed method

In this section, the principles of LLE are employed to conduct DR on the SPD matrix manifold, giving rise to the Lie-LLE algorithm. The SPD matrix manifold is highly suitable for LLE, this is because the tangent space of the SPD matrix manifold is a flat Euclidean space, allowing each tangent vector to be linearly represented by its neighboring tangent vectors, which aligns perfectly with the assumptions of LLE.

4.1. Tangent map

The manifold-manifold map and tangent map are two important concepts in differentiable manifolds, which realize the transformation between a manifold and its tangent space. Let $F : S_+^D \to S_+^d$ be a smooth map from the original manifold S_+^D to a new one S_+^d ($d \ll D$). The corresponding tangent map at a point $\boldsymbol{X}_1 \in S_+^D$ can be induced from F as

$$F_{*\boldsymbol{X}_1}(\log(\boldsymbol{X}_1)) = \boldsymbol{A}^T \log(\boldsymbol{X}_1)\boldsymbol{A}, \tag{10}$$

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where $\boldsymbol{X}_1 \in S_+^D$ and $\boldsymbol{A} \in R^{D\times d}$. To ensure that the mapped matrix is real and symmetric (i.e., lying in the tangent space S^d), $\boldsymbol{A}$ is required to have full column rank. It is easy to prove that the tangent map is an injection, thus can derive the manifold-manifold map F through the matrix exponential map by

$$\begin{aligned} F(\boldsymbol{X}_1) &= \exp\left(F_{*\boldsymbol{X}_1}(\log(\boldsymbol{X}_1))\right) \\ &= \exp(\boldsymbol{A}^T \log(\boldsymbol{X}_1)\boldsymbol{A}). \end{aligned} \tag{11}$$

Specifically, the tangent map on the SPD matrix manifold is learned by seeking a transformation matrix $\boldsymbol{A}$ that preserves the algebraic structure of S_+^D.

4.2. Lie-LLE

On the SPD matrix manifold, the geodesic distance based on LEM can be converted into the Euclidean distance in the tangent space at the identity matrix. Thus, we perform DR in the tangent space S^D and map the low-dimensional representation back to the manifold via the exponential map. More specifically, the DR process in the tangent space can be realized by learning a tangent map from S^D to S^d through seeking a transformation matrix $\boldsymbol{A}$. Given N SPD matrices $\boldsymbol{X}_i \in S_+^D (i = 1, \cdots, N)$, the specific steps for DR with Lie-LLE are given as follows:

- The adjacency graph construction

By Eq.4, the geodesic distance between any two SPD matrices $\boldsymbol{X}_i$ and $\boldsymbol{X}_j$ is calculated with

$$d(\boldsymbol{X}_i, \boldsymbol{X}_j) = \| \log(\boldsymbol{X}_i) - \log(\boldsymbol{X}_j)\|_F,$$

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where $\|\cdot\|_F$ denotes the Frobenius norm. For each point $\boldsymbol{X}_i$, its neighbors $\boldsymbol{X}_j (j \in \mathcal{O}(i))$ are identified by the k-nearest neighbor algorithm with LEM.

- Local linear reconstruction

In order to obtain the weight matrix which reconstructs each point from its neighbors, we need to minimize a cost function similar to Eq.5 in LLE as

$$f(\boldsymbol{W}) = \sum_{i=1}^{N} \left\| \log(\boldsymbol{X}_i) - \sum_{j \in \mathcal{O}(i)} \boldsymbol{W}_{ij} \log(\boldsymbol{X}_j) \right\|_F^2, \tag{12}$$

$$\text{s.t.} \sum_{j \in \mathcal{O}(i)} \boldsymbol{W}_{ij} = 1. \tag{13}$$

For each $\boldsymbol{X}_i$, the right of Eq.12 can be translated into

$$\begin{aligned}
&\left\| \log(\boldsymbol{X}_i) - \sum_{j \in \mathcal{O}(i)} \boldsymbol{W}_{ij} \log(\boldsymbol{X}_j) \right\|_F^2 \\
&= \operatorname{tr}\left((\log(\boldsymbol{X}_i) - \sum_{j \in \mathcal{O}(i)} \boldsymbol{W}_{ij} \log(\boldsymbol{X}_j))(\log(\boldsymbol{X}_i) - \sum_{j \in \mathcal{O}(i)} \boldsymbol{W}_{ij} \log(\boldsymbol{X}_j))^T \right) \\
&= \operatorname{tr}(\log^2(\boldsymbol{X}_i)) - 2 \operatorname{tr}(\sum_{j \in \mathcal{O}(i)} \boldsymbol{W}_{ij} \log(\boldsymbol{X}_i) \log(\boldsymbol{X}_j)) \\
&\quad + \operatorname{tr}(\sum_{m,n \in \mathcal{O}(i)} \boldsymbol{W}_{im} \boldsymbol{W}_{in} \log(\boldsymbol{X}_m) \log(\boldsymbol{X}_n)) \\
&= \operatorname{tr}(\log^2(\boldsymbol{X}_i)) - 2 \sum_{j \in \mathcal{O}(i)} \boldsymbol{W}_{ij} \operatorname{tr}(\log(\boldsymbol{X}_i) \log(\boldsymbol{X}_j)) \\
&\quad + \sum_{m,n \in \mathcal{O}(i)} \boldsymbol{W}_{im} \boldsymbol{W}_{in} \operatorname{tr}(\log(\boldsymbol{X}_m) \log(\boldsymbol{X}_n)) \\
&= \operatorname{tr}(\log^2(\boldsymbol{X}_i)) - 2 \boldsymbol{W}_i^T \boldsymbol{Q}_i + \boldsymbol{W}_i^T \boldsymbol{T}_i \boldsymbol{W}_i,
\end{aligned} \tag{14}$$

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where

$$\boldsymbol{Q}_i = \begin{pmatrix} \mathrm{tr}(\log(\boldsymbol{X}_i)\log(\boldsymbol{X}_1)) \\ \vdots \\ \mathrm{tr}(\log(\boldsymbol{X}_i)\log(\boldsymbol{X}_k)) \end{pmatrix}, \tag{15}$$

$$\boldsymbol{T}_i = \begin{pmatrix} \mathrm{tr}(\log(\boldsymbol{X}_1)\log(\boldsymbol{X}_1)) & \cdots & \mathrm{tr}(\log(\boldsymbol{X}_1)\log(\boldsymbol{X}_k)) \\ \vdots & \ddots & \vdots \\ \mathrm{tr}(\log(\boldsymbol{X}_k)\log(\boldsymbol{X}_1)) & \cdots & \mathrm{tr}(\log(\boldsymbol{X}_k)\log(\boldsymbol{X}_k)) \end{pmatrix}, \tag{16}$$

and $\boldsymbol{W}_i = (\boldsymbol{W}_{i1}, \cdots, \boldsymbol{W}_{ik})^T$ is the ith column of the weight matrix $\boldsymbol{W}$. Consequently, the minimization problem regarding $\boldsymbol{W}_i$ shown by Eq.12 and Eq.13 can be given as

$$\min f(\boldsymbol{W}_i) = \mathrm{tr}(\log^2(\boldsymbol{X}_i)) - 2\boldsymbol{W}_i^T\boldsymbol{Q}_i + \boldsymbol{W}_i^T\boldsymbol{T}_i\boldsymbol{W}_i, \tag{17}$$

$$\text{s. t. } \boldsymbol{W}_i^T\mathbf{1}_k = 1, \tag{18}$$

where $\mathbf{1}_k$ is a column vector of size $k \times 1$ in which all elements are ones. The Lagrange equation is introduced to solve the constrained optimization problem as

$$L(\boldsymbol{W}_i) = \mathrm{tr}(\log^2(\boldsymbol{X}_i)) - 2\boldsymbol{W}_i^T\boldsymbol{Q}_i + \boldsymbol{W}_i^T\boldsymbol{T}_i\boldsymbol{W}_i + \lambda(\boldsymbol{W}_i^T\mathbf{1}_k - 1). \tag{19}$$

In order to get the optimal solution, conducting derivative to $L(\boldsymbol{W}_i)$ with respect to $\boldsymbol{W}_i$ and set it to zero, i.e.,

$$\frac{\partial L(\boldsymbol{W}_i)}{\partial \boldsymbol{W}_i} = -2\boldsymbol{Q}_i + 2\boldsymbol{T}_i\boldsymbol{W}_i + \lambda\mathbf{1}_k = \mathbf{0}_k. \tag{20}$$

From Eq.20, the optimal solution is given by

$$\boldsymbol{W}_i = \boldsymbol{T}_i^{-1}(\boldsymbol{Q}_i - \frac{\lambda}{2}\mathbf{1}_k), \tag{21}$$

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which can be substituted into the constraint $\boldsymbol{W}_i^T\mathbf{1}_k = 1$ to find λ, i.e.,

$$(\boldsymbol{T}_i^{-1}(\boldsymbol{Q}_i - \frac{\lambda}{2}\mathbf{1}_k))^T\mathbf{1}_k = 1. \tag{22}$$

Through simple calculations, it can be obtained that

$$\lambda = \frac{2(\boldsymbol{Q}_i^T(\boldsymbol{T}_i^{-1})^T\mathbf{1}_k - 1)}{\mathbf{1}_k^T(\boldsymbol{T}_i^{-1})^T\mathbf{1}_k}, \tag{23}$$

where $\boldsymbol{Q}_i^T(\boldsymbol{T}_i^{-1})^T\mathbf{1}_k$ and $\mathbf{1}_k^T(\boldsymbol{T}_i^{-1})^T\mathbf{1}_k$ are both scalars. Finally by substituting λ back to Eq.21, the display expression of $\boldsymbol{W}_i$ is given as

$$\begin{aligned}\boldsymbol{W}_i =& \boldsymbol{T}_i^{-1}(\boldsymbol{Q}_i - \frac{\lambda}{2}\mathbf{1}_k) \\ =& \boldsymbol{T}_i^{-1}\big(\boldsymbol{Q}_i - \frac{\boldsymbol{Q}_i^T(\boldsymbol{T}_i^{-1})^T\mathbf{1}_k - 1}{\mathbf{1}_k^T(\boldsymbol{T}_i^{-1})^T\mathbf{1}_k}\mathbf{1}_k\big).\end{aligned} \tag{24}$$

- Low-dimensional embedding

In order to facilitate subsequent calculations, the weight matrix $\boldsymbol{W}_{k\times N}$ is extended to $\boldsymbol{W}_{N\times N}$. Specifically, $(\boldsymbol{W}_{N\times N})_{ij} = (\boldsymbol{W}_{k\times N})_{ij}$ if $j \in \mathcal{O}(i)$, otherwise $(\boldsymbol{W}_{N\times N})_{ij} = 0$. Then, the low-dimensional SPD matrices can be found by keeping the same reconstructing relationships of high-dimensional ones, which can be realized through minimizing a cost function similarly to Eq.7 with

$$\psi(\boldsymbol{Y}) = \sum_{i=1}^{N}\left\|\log(\boldsymbol{Y}_i) - \sum_{j=1}^{N}\boldsymbol{W}_{ij}\log(\boldsymbol{Y}_j)\right\|_F^2, \tag{25}$$

where $\boldsymbol{Y} = (\log(\boldsymbol{Y}_1), \cdots, \log(\boldsymbol{Y}_N))$ is the set of tangent vectors corresponding to the low-dimensional representations. Let $\boldsymbol{X} = (\log(\boldsymbol{X}_1), \cdots, \log(\boldsymbol{X}_N))$, according to Eq.10, the above problem can be represented as

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$$
\begin{aligned}
\psi(\boldsymbol{Y}) &= \sum_{i=1}^{N} \left\| \boldsymbol{A}^T \log(\boldsymbol{X}_i)\boldsymbol{A} - \sum_{j=1}^{N} \boldsymbol{W}_{ij}\boldsymbol{A}^T \log(\boldsymbol{X}_j)\boldsymbol{A} \right\|_F^2 \\
&= \sum_{i=1}^{N} \left\| \boldsymbol{A}^T \boldsymbol{X} \left(\begin{pmatrix} \boldsymbol{0}_{D\times D} \\ \vdots \\ \boldsymbol{I}_D \\ \vdots \\ \boldsymbol{0}_{D\times D} \end{pmatrix} - \begin{pmatrix} \boldsymbol{W}_{i1}\boldsymbol{I}_D \\ \vdots \\ \boldsymbol{W}_{ii}\boldsymbol{I}_D \\ \vdots \\ \boldsymbol{W}_{iN}\boldsymbol{I}_D \end{pmatrix} \right) \boldsymbol{A} \right\|_F^2 \\
&= \sum_{i=1}^{N} \operatorname{tr}\left(\boldsymbol{A}^T \boldsymbol{X} (\widetilde{\boldsymbol{E}}_i - \widetilde{\boldsymbol{W}}_i) \boldsymbol{A}\boldsymbol{A}^T (\widetilde{\boldsymbol{E}}_i - \widetilde{\boldsymbol{W}}_i)^T \boldsymbol{X}^T \boldsymbol{A} \right), \qquad (26)
\end{aligned}
$$

where

$$
\widetilde{\boldsymbol{E}}_i = \boldsymbol{e}_i \otimes \boldsymbol{I}_D = \left(\boldsymbol{0}_{D\times D}, \cdots, \boldsymbol{I}_D, \cdots, \boldsymbol{0}_{D\times D} \right)^T \qquad (27)
$$

and

$$
\widetilde{\boldsymbol{W}}_i = \boldsymbol{W}_i \otimes \boldsymbol{I}_D = \left(\boldsymbol{W}_{i1}\boldsymbol{I}_D, \cdots, \boldsymbol{W}_{iN}\boldsymbol{I}_D \right)^T \qquad (28)
$$

with $\boldsymbol{e}_i$ referring to the i-th column of the $N \times N$ identity matrix and $\otimes$ signifying the Kronecker product applied to matrices. To simplify Eq.26, it is necessary to introduce the property of the trace operator of matrix, which satisfies

$$
\operatorname{tr}(\boldsymbol{A}_1\boldsymbol{A}_2\boldsymbol{A}_3) = \operatorname{tr}(\boldsymbol{A}_2\boldsymbol{A}_3\boldsymbol{A}_1) = \operatorname{tr}(\boldsymbol{A}_3\boldsymbol{A}_1\boldsymbol{A}_2). \qquad (29)
$$

More specifically, if $\boldsymbol{A}_1$, $\boldsymbol{A}_2$ and $\boldsymbol{A}_3$ are symmetric square matrices with the same size, the trace of their product remains invariant not only under cyclic permutations but also under all permutations. It is easy to verify that

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$\boldsymbol{X}(\widetilde{\boldsymbol{E}}_i - \widetilde{\boldsymbol{W}}_i)$ is a symmetric matrix. Taking advantage of this property, the Eq.26 can be derived as

$$\begin{aligned}
&\sum_{i=1}^{N} \mathrm{tr}(\boldsymbol{A}^T\boldsymbol{X}(\widetilde{\boldsymbol{E}}_i - \widetilde{\boldsymbol{W}}_i)\boldsymbol{A}\boldsymbol{A}^T(\widetilde{\boldsymbol{E}}_i - \widetilde{\boldsymbol{W}}_i)^T\boldsymbol{X}^T\boldsymbol{A}) \\
=&\mathrm{tr}\big(\sum_{i=1}^{N} \boldsymbol{A}\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{X}(\widetilde{\boldsymbol{E}}_i - \widetilde{\boldsymbol{W}}_i)(\widetilde{\boldsymbol{E}}_i - \widetilde{\boldsymbol{W}}_i)^T\widetilde{\boldsymbol{X}}^T\big) \\
=&\mathrm{tr}\big(\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{X}(\sum_{i=1}^{N}(\widetilde{\boldsymbol{E}}_i - \widetilde{\boldsymbol{W}}_i)(\widetilde{\boldsymbol{E}}_i - \widetilde{\boldsymbol{W}}_i)^T)\boldsymbol{X}^T\big) \\
=&\mathrm{tr}(\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{X}\boldsymbol{P}\boldsymbol{X}^T),
\end{aligned}$$

where

$$\boldsymbol{P} = \sum_{i=1}^{N}(\widetilde{\boldsymbol{E}}_i - \widetilde{\boldsymbol{W}}_i)(\widetilde{\boldsymbol{E}}_i - \widetilde{\boldsymbol{W}}_i)^T. \tag{30}$$

Similar to LLE, the symmetric matrices need to be centered on the zero matrix and have unit covariance to obtain standardized data. The constraint conditions are respectively given by

$$\sum_{i=1}^{N}\log(\boldsymbol{Y}_i) = \boldsymbol{0}_{d\times d}, \boldsymbol{Y}\boldsymbol{Y}^T = N\boldsymbol{I}_d. \tag{31}$$

The Lagrange multiplier method is then employed to solve the constraint optimization problem with

$$L(\boldsymbol{A}) = \mathrm{tr}(\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{X}\boldsymbol{P}\boldsymbol{X}^T) - \lambda\,\mathrm{tr}(\boldsymbol{Y}\boldsymbol{Y}^T - N\boldsymbol{I}_d), \tag{32}$$

where

$$\begin{aligned}
\mathrm{tr}(\boldsymbol{Y}\boldsymbol{Y}^T - N\boldsymbol{I}_d) =& \mathrm{tr}\big((\log(\boldsymbol{Y}_1),\cdots,\log(\boldsymbol{Y}_N))(\log(\boldsymbol{Y}_1),\cdots,\log(\boldsymbol{Y}_N))^T - N\boldsymbol{I}_d\big) \\
=&\mathrm{tr}(\sum_{i=1}^{N}\boldsymbol{A}^T\log(\boldsymbol{X}_i)\boldsymbol{A}\boldsymbol{A}^T\log(\boldsymbol{X}_i)\boldsymbol{A} - N\boldsymbol{I}_d) \\
=&\mathrm{tr}(\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{X}\boldsymbol{X}^T) - Nd.
\end{aligned}$$

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Substituting the above equation into Eq.32, we have

$$L(\boldsymbol{A}) = \mathrm{tr}(\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{X}\boldsymbol{P}\boldsymbol{X}^T) - \lambda\,\mathrm{tr}(\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{X}\boldsymbol{X}^T) + \lambda N d. \tag{33}$$

In order to get the optimal solution, we can conduct the derivative of Eq.33 with respect to A, resulting in

$$\begin{aligned}\frac{\partial L(\boldsymbol{A})}{\partial \boldsymbol{A}} =& 4(\mathrm{tr}(\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{X}\boldsymbol{P}\boldsymbol{X}^T) - \lambda\,\mathrm{tr}(\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{X}\boldsymbol{X}^T)) \\ =& 4\,\mathrm{tr}(\boldsymbol{A}^T\boldsymbol{A}(\boldsymbol{A}^T\boldsymbol{X}\boldsymbol{P}\boldsymbol{X}^T - \lambda\boldsymbol{A}^T\boldsymbol{X}\boldsymbol{X}^T)).\end{aligned} \tag{34}$$

Since $\boldsymbol{A}$ is a $D \times d$ full rank matrix, $\boldsymbol{A}^T\boldsymbol{A}$ is a $D \times D$ SPD matrix. To obtain $\frac{\partial L(\boldsymbol{A})}{\partial \boldsymbol{A}} = 0$, it should solve the following generalized eigenfunction problem

$$\boldsymbol{X}\boldsymbol{P}^T\boldsymbol{X}^T\boldsymbol{A} = \lambda\boldsymbol{X}\boldsymbol{X}^T\boldsymbol{A}. \tag{35}$$

After obtaining the d smallest non-zero eigenvalues $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_d$ and the corresponding eigenvectors $\boldsymbol{a}_1, \boldsymbol{a}_2, \cdots, \boldsymbol{a}_d$, the mapping matrix $\boldsymbol{A}$ can be represented as $\boldsymbol{A} = [\boldsymbol{a}_1, \boldsymbol{a}_2, \cdots, \boldsymbol{a}_d]$. For $\boldsymbol{X}_i \in S_+^D$, the corresponding low-dimensional matrix $\boldsymbol{Y}_i \in S_+^d$ can be expressed by $\boldsymbol{Y}_i = \exp(\boldsymbol{A}^T \log(\boldsymbol{X}_i)\boldsymbol{A})$.

4.3. Complexity analysis

The main time-consuming aspect of Lie-LLE lies in the computation of the weight matrix $\boldsymbol{W} = [\boldsymbol{W}_1, \cdots, \boldsymbol{W}_N]$. As can be seen in Eq.24, $\boldsymbol{W}_i$ is calculated by $\boldsymbol{Q}_i$ and $\boldsymbol{T}_i$. Specifically, the computational complexity for calculating $\boldsymbol{Q}_i$ in Eq.15 is $O(k)$ and the computing for $\boldsymbol{T}_i$ in Eq.16 is $O(k^2D^3)$, where k is the number of k-nearest neighbors and D is the size of the high-dimensional SPD matrix. By combining these components, the total complexity for each $\boldsymbol{W}_i$ becomes $O(3k^2+3k)$. Consequently, the overall complexity for constructing the full weight matrix $\boldsymbol{W}$ is $O(N(k^2D^3+3k^2+4k))$, where N is the total number of training samples.

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The main pseudo-code of the proposed method is given by Algorithm 1.

Algorithm 1 LLE on SPD matrix Lie group (Lie-LLE)

Input: A set of SPD matrices $\{\boldsymbol{X}_1, \cdots, \boldsymbol{X}_N\}$, $\boldsymbol{X}_i \in S_+^D$, the number of neighbors: k, target size of the low-dimensional SPD matrix: d.

Output: The low-dimensional SPD matrices $\{\boldsymbol{Y}_1, \cdots, \boldsymbol{Y}_N\}$,$\boldsymbol{Y}_i \in S_+^d$.

1. Find the k nearest neighbors of $\boldsymbol{X}_i$ by LEM.
2. **for** $i = 1$ to N **do**

   Calculate $\boldsymbol{Q}_i$, $\boldsymbol{T}_i$ and $\boldsymbol{W}_i$ by Eq.15, Eq.16 and Eq.24

   end for
3. $\boldsymbol{W}_{k\times N} \leftarrow [\boldsymbol{W}_1, \cdots, \boldsymbol{W}_N]$, extend the weight matrix $\boldsymbol{W}_{k\times N}$ to $\boldsymbol{W}_{N\times N}$.
4. $\boldsymbol{P} \leftarrow \boldsymbol{0}_{ND\times ND}$

   for $i = 1$ to N **do**

   $\widetilde{\boldsymbol{W}}_i \leftarrow \boldsymbol{W}_i \otimes \boldsymbol{I}_D$, $\widetilde{\boldsymbol{E}}_i \leftarrow \boldsymbol{e}_i \otimes \boldsymbol{I}_D$

   $\boldsymbol{P} \leftarrow \boldsymbol{P} + (\widetilde{\boldsymbol{E}}_i - \widetilde{\boldsymbol{W}}_i)(\widetilde{\boldsymbol{E}}_i - \widetilde{\boldsymbol{W}}_i)^T$

   end for
5. $\boldsymbol{X} \leftarrow [\log(\boldsymbol{X}_1), \cdots, \log(\boldsymbol{X}_N)]$
6. $[\boldsymbol{V}_c, \boldsymbol{V}_l] = \mathrm{eig}(\boldsymbol{X}\boldsymbol{P}^T\boldsymbol{X}^T, \boldsymbol{X}\boldsymbol{X}^T)$

   $\boldsymbol{H} \leftarrow [\mathrm{diag}(\boldsymbol{V}_l), (\boldsymbol{V}_c)^T]$, $\boldsymbol{J} \leftarrow \mathrm{sortrows}(\boldsymbol{H})$

   if $\boldsymbol{J}(1,1) == 0$

   $\boldsymbol{A}^T \leftarrow \boldsymbol{J}(2 : d+1, 2 : end)$

   else

   $\boldsymbol{A}^T \leftarrow \boldsymbol{J}(1 : d, 2 : end)$

   end if
8. $\boldsymbol{Y}_i = \exp(\boldsymbol{A}^T \log(\boldsymbol{X}_i)\boldsymbol{A})$.

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5. Experiments

In this section, to validate the effectiveness of the proposed method, we first perform visualization experiments on artificial generated datasets. Subsequently, image set classification are performed on Honda (UCSD) [31] and ETH-80 [32] database.

5.1. Comparative methods and settings

In this paper, PML[33], CDL[15], LEML[17], ($(2D)^2$PCA)[22], Lie-LPP[23] and RMTSLA-SPDDR[19], are compared with our method. In the experiments, the parameters of these algorithms are selected as follows: For LEML and RMTSLA-SPDDR, several sensitive parameters are selected based on the empirical recommendations provided by the author. For CDL, the LDA-based version is applied, which adopts the same setting as [15] with $c-1$ as target dimension (c is the number of image sets classes in training process). For PML, the parameters are kept the same as [33] and the dimensionality of linear subspace is determined by cross-validation. For Lie-LPP, the number of k-nearest neighbors is tuned in the range of 5 to 10, and the t in the Gaussian kernel is set in the range of 0.1 to 10. For the proposed method Lie-LLE, the number of nearest neighbors k is selected from 5 to 8. The low-dimensionality SPD matrices in Lie-LLE, Lie-LPP, $(2D)^2$PCA and RMTSLA-SPDDR are all of size $d \times d$, which have dimension of $d(d+1)/2$. The d is selected different value in each experiment and is mentioned respectively in the following text.

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5.2. Visualization analysis of DR with different methods

To visualize the intrinsic geometric structure of the SPD matrix manifold, we conduct DR on two different sets of SPD matrices and represent them in $3D$ space. The DR results of the two dataset are given by Lie-LLE, Lie-LPP, $(2D)^2$PCA and RMTSLA-SPDDR, which can directly obtain low-dimensional SPD matrices for visualization.

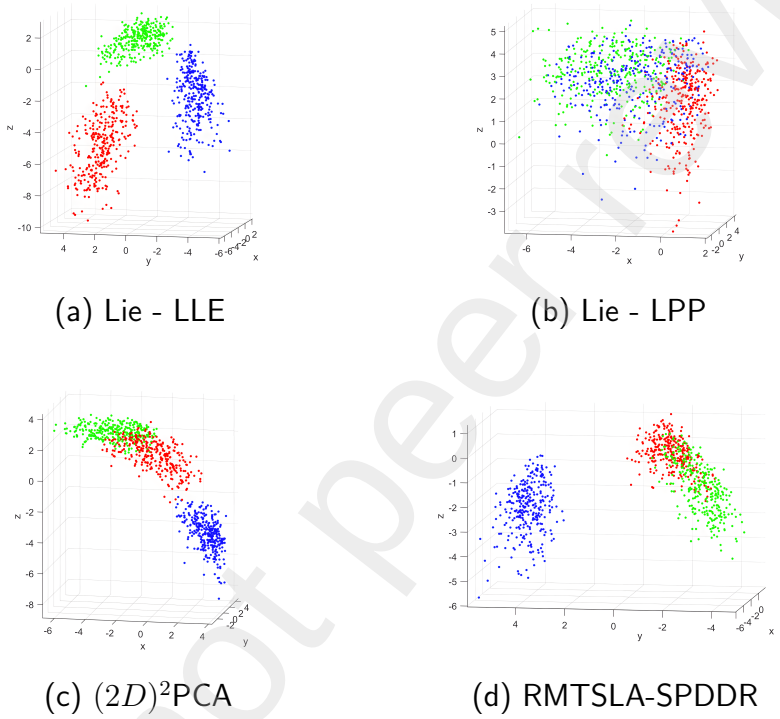
For the first dataset, we first generate three reference SPD matrices of size 3×3, ensuring pairwise Log-Euclidean distances range from 10 to 15. Then, 300 additional 3×3 SPD matrices are randomly generated around each reference matrix within a Log-Euclidean distance of 4.8. For the sake of simplicity and uniformity, set all diagonal entries of these matrices to 1. All these 900 SPD matrices are reduced to low-dimensional ones of size 2×2 (i.e., $d = 2$) using different DR methods. The results are represented in the logarithmic domain, as shown in Fig. 1 (a)-(d). It is evident that the embedded points of Lie-LLE exhibit clear cluster boundaries, which demonstrates its superior performance in preserving the geometric relationships of high-dimensional SPD matrices.

For the second dataset, we randomly construct three clusters of 4×4 SPD matrices with determinants of 0.02, 20, and 2000, and each cluster consists of 300 SPD matrices. The dimension of these matrices is 10 and their logarithm representation lie on a 9-dimensional parallel hyperplane in the tangent space, a geometric structure that is difficult to visualize. Using DR algorithms, the inherent structure of these matrices can be effectively projected and represented within the space of 2×2 symmetric matrices as illustrated in Fig. 2 (a)-(d). It is evident that Lie-LLE performs better in maintaining

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distinct boundaries for these matrices compared to other algorithms, which suggests its good ability to distinguish matrices with significantly different determinants.



(a) Lie - LLE (b) Lie - LPP

(c) $(2D)^2$PCA (d) RMTSLA-SPDDR

Fig. 1: Visualization of three clusters of 3×3 SPD matrices with mutually disjoint Log-Euclidean distances, reduced to 2×2 matrices via Lie-LLE, Lie-LPP, $(2D)^2$PCA and RMTSLA-SPDDR in the tangent space.

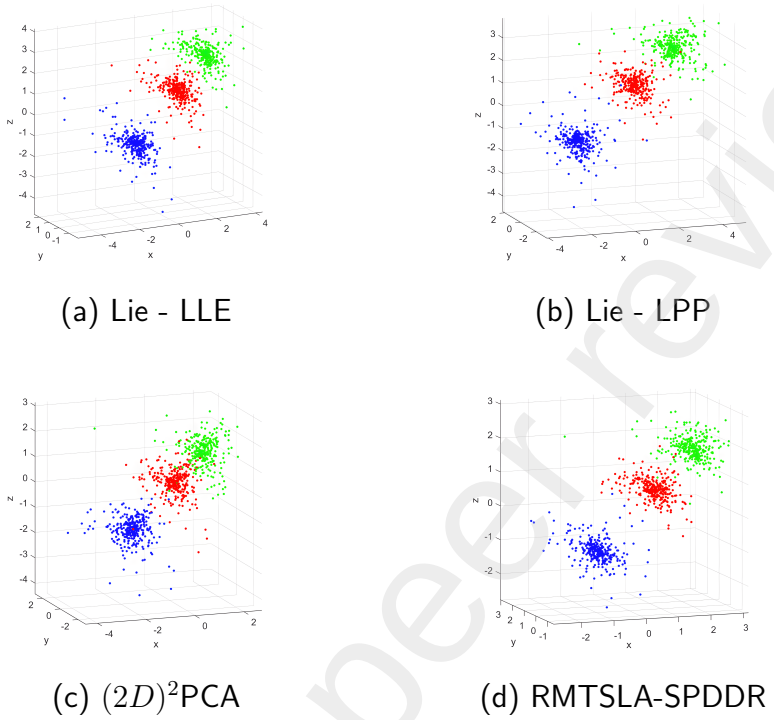
5.3. Video-based face recognition

5.3.1. Database and modeling

This paper utilizes the Honda database for video-based face recognition, which consist of 59 videos from 20 individuals. A cascaded face detector is

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(a) Lie - LLE (b) Lie - LPP

(c) $(2D)^2$PCA (d) RMTSLA-SPDDR

Fig. 2: Visualization of three clusters of 4×4 SPD matrices with determinants 0.02, 20 and 2000 respectively, reduced to 2×2 matrices via Lie-LLE, Lie-LPP, $(2D)^2$PCA and RMTSLA-SPDDR in the tangent space.

employed to extract face images from the videos, and the images are resized to 20×20 pixels. In order to eliminate the impact of lighting variations, histogram equalization is applied for preprocessing. Each video generates a set of face images, and each person has $2 \sim 5$ image sets. In each experiment, one image set per person is selected for training, while the remaining sets are used for testing. Specifically, a single training image set is randomly divided into two disjoint subsets to construct intra-class similarity. The d is tuned

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60 in this database. All experiments are performed ten times.

Covariance matrix is used for constructing the original SPD matrix of Honda database. Suppose there are N pictures in an image set: $\boldsymbol{S} = \{\boldsymbol{s}_1, \boldsymbol{s}_2, \cdots, \boldsymbol{s}_N\}, \boldsymbol{s}_i \in R^D$ is the feature vector of the ith image (e.g., the raw intensity feature vector). The covariance matrix of the image set can be represented as

$$\boldsymbol{C} = \frac{1}{N-1}\sum_{i=1}^{N}(\boldsymbol{s}_i - \boldsymbol{m})(\boldsymbol{s}_i - \boldsymbol{m})^T, \tag{36}$$

where $\boldsymbol{m} = \frac{1}{N}\sum_{i=1}^{N}\boldsymbol{s}_i$ represents the mean feature of the image set. To ensure that the covariance matrix is positive definite, $\delta\boldsymbol{I}_D$ can be added to the matrix when the number of samples N is smaller than the dimension of feature vectors D. Here, δ is defined as $10^{-3} \times \mathrm{tr}(\boldsymbol{C})$.

5.3.2. Parametric analysis

To explore the impact of the two important parameters in Lie-LLE, we conduct an analysis of the results on the Honda database as shown in Fig. 3, which illustrates the mean accuracy for different values of k across various target size d.

Notably, the performance is suboptimal when each point considers only the same class points as its nearest neighbors (i.e., $k = 1$), and it also deteriorates when too many neighbors are included (e.g., $k = 15$). As a result, small k values fail to take into account the relationships between different class points and may separate points of the same class, resulting in an inaccurate construction of the adjacency graph. On the other hand, too large k values neglect the local relationships of data points and can introduce noise through inclusion of non-relevant points. Therefore, optimal k values are

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essential to preserve topological integrity in manifold embedding by striking a balance between local and global structures.

The results in Fig. 3 also demonstrate a non-monotonic relationship between the target dimension and classification accuracy when fixing the neighborhood size k. Specifically, at a very low target dimension with a small target size d, the results is suboptimal due to a significant information loss caused by over-constrained embeddings. As the dimension increases (i.e., d increases), the accuracy gradually improves, reaching a peak when the target dimension aligns with the intrinsic geometric complexity of the data. However, beyond this range, further increasing dimension introduces redundant noise components, leading to a gradual decline in accuracy as the model over fits to irrelevant geometric details. Notably, at sufficiently high dimensions with a large d, the accuracy for different k values converge to a stable baseline, regardless of neighborhood size. This convergence arises because high-dimensional embeddings asymptotically approximate the global principal components of the data, effectively neutralizing the impact of local neighborhood selection. These observations highlight the necessity of selecting target dimensions based on prior knowledge of manifold complexity to achieve the most effective DR results.

5.3.3. Comparison of DR and Non-DR

To explore the necessity of DR, we compare the recognition results performed on the original SPD matrices with those performed on low-dimensional ones after applying DR techniques, as shown in Table 1. For clarity, recognition on the original manifold with Euclidean metric and LEM is referred to as EM-NonDR and LEM-NonDR, respectively. The results in Table 1

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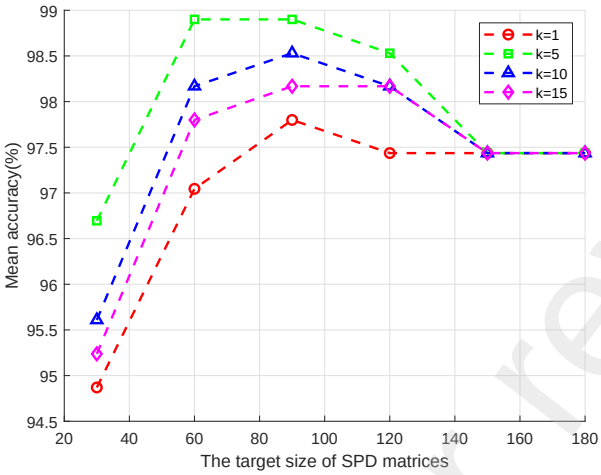


Fig. 3: The mean accuracy with different k nearest neighbors across various target size d of Lie-LLE on Honda database.

show that the Non-DR methods yield suboptimal outcomes than the DR methods. This indicates that the high-dimensional matrices often contain a significant amount of redundant information, and the DR techniques can improve recognition accuracy by extracting effective information. Additionally, the comparison between EM-NonDR and LEM-NonDR methods shows that LEM can better reflect the geometric structure of the SPD matrix manifold than Euclidean metric, resulting in improved performance.

5.3.4. Comparison of different DR algorithms

Table 1 presents the mean classification accuracy and standard deviation of various methods across 10 random tests. It is evident that Lie-LLE outperforms CDL, as Lie-LLE operates within the LEM framework, which respects the structure of the SPD matrix manifold, while CDL relies on a kernel method that overlooks this structure. In comparison to PML, which

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Table 1: Image set classification results of advanced methods on Honda database

Method	LEM-NonDR	EM-NonDR	CDL	PML
Accuracies	97.44 ± 0.01	87.95 ± 3.96	97.69 ± 1.46	96.67 ± 1.24
Method	Lie-LPP	$(2D)^2$PCA	RMTSLA-SPDDR	**Lie-LLE**
Accuracies	97.95 ± 1.08	98.20 ± 1.24	95.38 ± 1.08	$\mathbf{98.97 \pm 1.32}$

calculates the distance between low-dimensional matrices by learning a projection on the Grassmann manifold, it can be concluded that conducting DR on the SPD matrix manifold by modeling image sets as SPD matrices yields better results than conducting DR on the Grassmann manifold by modeling image sets as linear subspaces. Similar to Lie-LLE, $(2D)^2$PCA, Lie-LPP and RMTSLA-SPDDR all learn a linear mapping matrix to generate low-dimensional SPD matrices. Fig. 4 illustrates the mean accuracy of different DR methods across varying target size d. It can be observed that Lie-LLE maintains consistent high accuracy across all test d, indicating its ability to derive more effective projection matrix than other three methods.

5.4. Set-based object categorization

5.4.1. Database and modeling

For the set-based object categorization task, experiments are conducted on the ETH-80 database, which is a typical database of object classification. It contains eight categories. Each category has ten image sets, with 41 images taken from different angles. Each image is resized to 20×20 pixels. Five image sets per subject are randomly selected for training and the remaining

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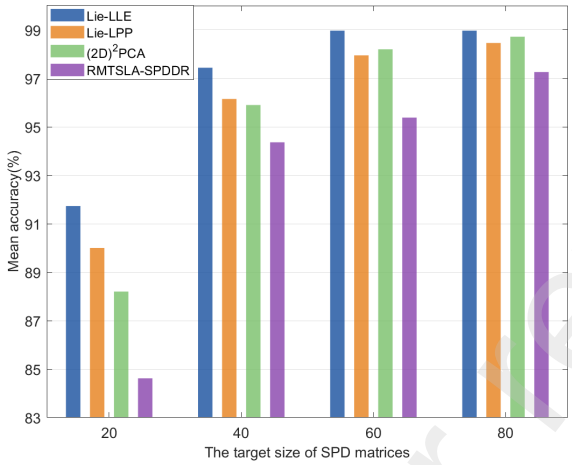


Fig. 4: Image set classification results of Lie-LLE, Lie-LPP, $(2D)^2$PCA and RMTSLA-SPDDR across various target size d on Honda database.

sets are used for testing. d is chosen 30 in this database. All experiments are performed ten times.

The original SPD matrix of ETH-80 database is based on Gaussian model matrix. Assume the Gaussian model for each image set is $N(\boldsymbol{m}, \boldsymbol{C})(\boldsymbol{m} \in R^D$ is mean and $\boldsymbol{C} \in S_+^D$ is the covariance matrix). After employing information geometry theory [34], the Gaussian model $N(\boldsymbol{m}, \boldsymbol{C})$ can be transformed to an SPD matrix of size $(D+1) \times (D+1)$ as

$$|\boldsymbol{C}|^{-\frac{1}{D+1}} \begin{bmatrix} \boldsymbol{C} + \boldsymbol{m}\boldsymbol{m}^T & \boldsymbol{m} \\ \boldsymbol{m}^T & 1 \end{bmatrix}. \tag{37}$$

For detailed information about Gaussian model refer to [35]. Similarly, $\delta \boldsymbol{I}_D$ is added to the matrix to keep its positive definiteness.

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5.4.2. Comparison of DR and Non-DR

Table2 shows the classification results of Non-DR methods and DR methods, which further justifies the necessity of DR and highlights the advantages of LEM on the SPD matrix manifold.

5.4.3. Comparison of different DR algorithms

Table2 presents the classification performance comparison of different methods across 10 random tests on ETH-80 database. It is evident that the proposed method demonstrates superior accuracy and stability compared to other methods. Table2 shows that LEM exhibits a poorer ability to learn a linear mapping on the SPD matrix manifold compared to Lie-LLE. Since Lie-LLE, Lie-LPP, $(2D)^2$PCA and RMTSLA-SPDDR all preserve the SPD matrix structure in reduced dimensions, their performance across varying target size d is analyzed in Fig. 5. Lie-LLE maintains consistent high accuracy across all tested d, attributed to its theoretical foundation rooted in LLE which endows it with superior capabilities in modeling nonlinear data structures.

Table 2: Image set classification results of advanced methods on ETH-80 database

Method	LEM-NonDR	EM-NonDR	PML	LEML
Accuracies	88.50 ± 2.93	62.75 ± 5.95	86.94 ± 4.30	88.50 ± 2.93
Method	$(2D)^2$PCA	Lie-LPP	RMTSLA-SPDDR	**Lie-LLE**
Accuracies	90.00 ± 3.12	90.50 ± 3.29	88.75 ± 3.17	$\mathbf{90.75 \pm 2.65}$

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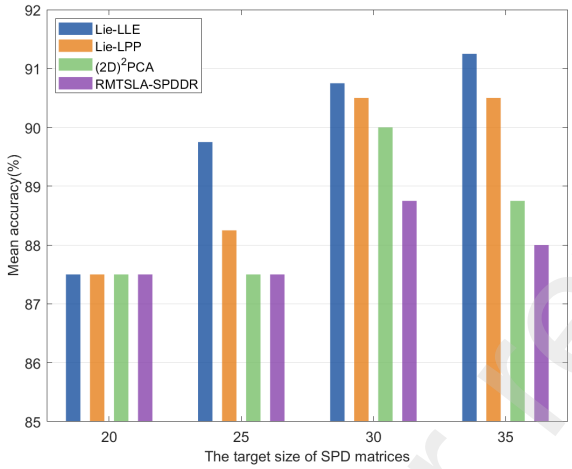


Fig. 5: Image set classification results of Lie-LLE, Lie-LPP, $(2D)^2$PCA and RMTSLA-SPDDR across various target size d on ETH-80 database.

6. Conclusions

In this paper, we introduce a novel DR algorithm for SPD matrix manifold, termed Lie-LLE. This method recovers the latent low-dimensional manifold structure embedded in high-dimensional SPD matrices by leveraging local linear geometric dependencies between matrix samples. Notably, Lie-LLE entails merely two tunable parameters, which are readily adjustable to achieve optimal performance across diverse datasets. Moreover, the algorithm inherently preserves the intrinsic geometric structures of SPD matrices, while demonstrating superior generalization performance on new samples. Lie-LLE is applicable to a wide range of machine learning tasks employing SPD matrices as feature descriptors–including but not limited to computer vision, medical imaging and signal processing. By constructing an embedded low-dimensional manifold, Lie-LLE effectively minimizes computational

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complexity, reduces storage requirements and enhances classification accuracy, establishing itself as a robust tool for SPD matrix-based data analysis.

Although Lie-LLE has many advantages, it also has some limitations. For instance, it has computational burden from matrix logarithmic operations and the determination of the weight matrix, which restrict its applicability to high-dimensional scenarios. In the future, we intend to explore preprocessing techniques (e.g., PCA or random projection) with Lie-LLE to mitigate computational complexity, thereby efficiently processing high-dimensional inputs. Additionally, we will conduct in-depth research on extending more classical manifold learning algorithms to the SPD matrix manifold and other Riemannian manifolds, such as the Glassmann manifold. Through these efforts, we hope to expand the manifold learning framework and broaden its application scenarios.

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